

(a) Problem setup



First-visit visual evidence
road-sign crop written once

(b) Memory write/store

crop path

feature vector

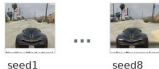
metadata

External memory

Inference-time only

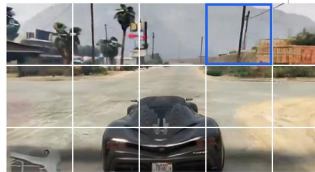
(c) Candidate generation

Frozen MatrixGame-2



candidate seed1 ... seed8
same pool before reranking

(d) Memory read by patch matching



Regular grid patches
Cosine similarity

Feature encoder
best local patch

(e) Candidate reranking



No memory: seed1



With memory: seed8

seed1 -> seed8
selected output changes

No object segmentation. Patches are regular grid cells.

ResNet / DINO / CLIP are feature extractors.